

A computability boundary in complete quantum information comparison

Coherent-reference localization versus finite inverse-action compression

Astra independent research campaign

Internally reconstructed proofs; external review and priority unresolved

11 October 2026

Abstract

The strongest result developed in this campaign is a structural classification, not a new capacity or Hamiltonian undecidability claim. For finite-dimensional channels, complete dominance for any single finite-at-zero operator-convex standard Petz divergence with a nonzero interior resolvent component is equivalent to complete SLD dominance. A coherent auxiliary reference localizes the generator's representing measure at one chosen support point. At the pure quadratic endpoint, a different elementary operation compresses every violation: reference dimension at most $1 + b^2$ suffices when the target channel has effective output dimension b , improving to $1 + v$ for a target POVM with v outcomes. Consequently quadratic comparison is exactly decidable on rational POVMs. These two proofs do not require Dots' nonlocal-game reduction. If that reduction is valid, they give a sharp decidable/coRE-complete boundary, including Petz Rényi parameter 2 versus every fixed parameter in $(0, 2)$. The manuscript received targeted adversarial reconstruction, and one complex coordinate convention needs correction. Its hardness is retained as a named dependency, not promoted to an independently certified theorem. No result here establishes historic 9–10/10 significance.

Read first. The complete mathematical arguments are in Sections 1–3. Section 4 records the source reduction's exact status and correction; Sections 5–7 give the consequences, falsification results, and research decision. Every claim is either proved below, explicitly inherited, or marked conditional/unresolved. The proofs were developed and cross-checked by AI agents in this campaign. Agreement, compilation, and finite numerical checks are not external or formal certification. The attached evidence archive preserves the independent investigations, diagnostics, literature receipts, and unchanged source PDF.

Computational interface. An input measurement channel is a finite list of rational complex Hermitian effects summing to the identity. “Strictly positive” means each effect is positive definite. Physical dimension and outcome count may grow with the encoded machine. References are arbitrary finite-dimensional untouched quantum systems, and their states/matrices are universally quantified, not supplied as bounded auxiliary input. State pairs may have unequal reference marginals. Exact decision and fixed-additive numerical approximation are distinct tasks. No polynomial-time construction is asserted when a succinct nonlocal game is expanded to its explicit rational table.

1 Definitions and fixed-generator universality

For positive matrices define $\Omega_A(X) = (AX + XA)/2$ and $g_A(K) = \text{Tr } K\Omega_A^{-1}(K)$ for Hermitian K . Channels have the same finite-dimensional input. Their output matrices are restricted to the

support of the channel's output on the identity. A faithful input gives a positive definite output on that support. Complete SLD dominance means

$$g_{P_R(\rho)}(P_R(H)) \geq g_{Q_R(\rho)}(Q_R(H))$$

for every finite untouched R , faithful input ρ and traceless Hermitian H . Complete divergence dominance means comparison on every faithful pair and every finite untouched reference. No equality of reference marginals is imposed.

The standard modular Petz divergence has spectral definition

$$D_f(A\|B) = \sum_{i,j} b_j f(a_i/b_j) |\langle u_i, v_j \rangle|^2.$$

For $s > 0$ put $f_s(x) = (x-1)^2/(x+s)$.

Theorem 1 (Fixed-generator universality). *Suppose an operator-convex generator has the representation*

$$f(x) = a + b(x-1) + c(x-1)^2 + \int_{(0,\infty)} f_s(x) d\nu(s), \quad c \geq 0,$$

where ν is positive, nonzero, and $\int (1+s)^{-1} d\nu(s) < \infty$ and $\int_{(0,1]} s^{-1} d\nu(s) < \infty$. Then complete D_f dominance is equivalent to complete SLD dominance. In particular this applies to every finite-at-zero operator-convex generator whose representation has nonzero interior resolvent mass. Affine terms are immaterial on state pairs. The pure quadratic case is treated separately in Theorem 3.

Forward implication. Fix faithful ρ, σ including any previous reference and put $X = \rho - \sigma$. Append an untouched qubit and use

$$\Theta_s = \frac{\text{diag}(\rho, s\sigma)}{1+s}, \quad H_X = \begin{pmatrix} 0 & X \\ X & 0 \end{pmatrix}.$$

The off-diagonal Sylvester equation gives exactly

$$g_{\Theta_s}(H_X) = 4(1+s) \langle X, (L_\rho + sR_\sigma)^{-1} X \rangle = 4(1+s) D_{f_s}(\rho\|\sigma).$$

The same identity holds after either channel. Thus complete SLD dominance compares every D_{f_s} . Taking $s \rightarrow \infty$ in sD_{f_s} compares $D_{(x-1)^2}$. Integrating the nonnegative generator representation compares D_f . Affine terms give equal constants because both arguments have trace one. All limits here are finite-dimensional and on faithful pairs. $\square$

1.1 The coherent-reference localization mechanism

Lemma 1 (Exact saturated denominator Hessian). *Fix a faithful state A , a Hermitian matrix K , positive $w_0, \lambda_1, \dots, \lambda_N, w_*$, leaf vectors v, w with $\|v\| = 1$, and $0 < p < 1$. On centre, leaves, and a decoupled dummy coordinate let*

$$W = \text{diag}(w_0, \lambda_1, \dots, \lambda_N, w_*), \quad Z = \text{Tr } W, \quad L = e_0 w^* + w e_0^*,$$

$$\tau_p = p v v^* + (1-p) e_* e_*^*, \quad F(t) = w^* (W_{\text{leaf}} + tI)^{-1} v.$$

Write $A = \sum_i a_i |i\rangle\langle i|$. Then

$$\begin{aligned} & \frac{d^2}{d\epsilon^2} D_{f_s} \left(A \otimes \tau_p \left\| \frac{A \otimes W + \epsilon K \otimes L}{Z} \right\| \right) \Big|_0 \\ &= 2p^2 Z \frac{(1+s)^2}{s^3} |F(pZ/s)|^2 \sum_{i,j} \frac{|K_{ij}|^2}{a_i p Z/s + a_j w_0}. \end{aligned} \tag{H}$$

For $f(x) = (x-1)^2$ the same second derivative is zero whenever $F(0) = 0$.

Proof. For a fixed numerator with positive eigenvalues r and associated eigenvectors u_r , division of the scalar perspective gives

$$D_{f_s}(A_0\|B) = \frac{\text{Tr } B}{s} - \frac{2s+1}{s^2} \text{Tr } A_0 + \sum_{r>0} \frac{r^2(s+1)^2}{s^2} \langle u_r, (r+sB)^{-1} u_r \rangle.$$

The zero numerator eigenspaces are already included in the first term. With $Y = K \otimes L/Z$, twice differentiating the inverse produces $2s^2(r+sB)^{-1}Y(r+sB)^{-1}Y(r+sB)^{-1}$. Only the numerator eigenvectors $|i\rangle \otimes v$, with $r = pa_i$, contribute: L is zero on the dummy. At $\epsilon = 0$, the two exterior resolvents give

$$\left| \sum_k \frac{\overline{w_k} v_k}{pa_i + sa_i \lambda_k / Z} \right|^2 = \frac{Z^2}{s^2 a_i^2} |F(pZ/s)|^2.$$

The centre resolvent on system index j is $\frac{Z}{s}(a_i pZ/s + a_j w_0)^{-1}$. Multiplying these factors gives (H). The quadratic divergence is $\text{Tr } A_0^2 B^{-1} - 1$; its corresponding exterior resolvent sum is proportional to $F(0)$. Hence its Hessian vanishes when $F(0) = 0$. $\square$

Reverse implication in Theorem 1. Fix the original faithful input ρ and tangent H , including any old reference. For either channel write $A = \Phi(\rho)$, $K = \Phi(H)$. Append the auxiliary matrices in Lemma 1 at the input. At $\epsilon = 0$,

$$D_f(A \otimes \tau_p \| A \otimes W/Z) = D_f(\tau_p \| W/Z),$$

independently of the channel. The denominator is faithful for sufficiently small positive and negative ϵ . Although the numerator has a kernel, the hypothesis extends to it by replacing τ_p with a faithful mixture and taking the mixture parameter to zero. This is valid because $f(0+) < \infty$ and the denominator stays faithful. The source-minus-target difference is therefore nonnegative near zero and zero at zero. Its second derivative is nonnegative.

Choose $s_0 \in \text{supp } \nu \cap (0, \infty)$, fix $w_0 > 0$, and put $c_0 = s_0 w_0$. For sufficiently large integers m , use $N = 2m + 1$ leaves with

$$\lambda_j = w_0(1 + j/m^3), \quad 0 \leq j \leq 2m, \quad F_m(t) = \frac{t^m}{\prod_{j=0}^{2m} (t + \lambda_j)}.$$

Partial fractions realize F_m as a finite simple-resolvent sum, with v uniform and w chosen from the coefficients. Set $w_* = w_0$, $Z_m = \text{Tr } W_m$ and $p_m = c_0/Z_m < 1$. Thus $p_m Z_m = c_0$ is independent of m . Since $m \geq 1$, $F_m(0) = 0$ kills the quadratic Hessian. Integrating (H) and discarding its common positive factor, the nonnegative Hessian difference compares

$$\sum_{i,j} |K_{ij}|^2 \int_{(0,\infty)} \frac{(1+s)^2}{s^3} \frac{|F_m(c_0/s)|^2}{a_i c_0/s + a_j w_0} d\nu(s). \quad (1)$$

Differentiation under the representing integral is legitimate. For each fixed finite instance, every positive numerator eigenvalue r is bounded away from zero, and the denominator is bounded below on a neighborhood of zero. The preceding resolvent formula bounds the absolute second derivative by $C/(1+s)$, with a constant depending on that finite instance. The representation's integrability condition supplies domination.

Here is the precise concentration argument, including singular measures. Let $t = c_0/s$, and push the finite positive measure

$$\frac{(1+s)^2}{s^3(t+w_0)^2} d\nu(s)$$

to a measure μ on $(0, \infty)$. It is finite: its weight is $O(s^{-1})$ both as $s \rightarrow 0$ and $s \rightarrow \infty$, and the hypotheses on ν give both integrability bounds. The point w_0 belongs to its support because s_0 belongs to the support of ν . Put

$$h(t) = \frac{4w_0 t}{(t+w_0)^2}.$$

Then $0 < h \leq 1$, with the unique maximum at w_0 , and

$$f_m(t) = \frac{t^m}{(t + w_0)^{2m+1}} = \frac{h(t)^m}{(4w_0)^m(t + w_0)}, \quad e^{-2/m-1/m^2} \leq F_m(t)/f_m(t) \leq 1.$$

The probability measures proportional to $h(t)^{2m}d\mu(t)$ concentrate at w_0 . Indeed, outside any fixed neighborhood their exponential base is at most $r < 1$, whereas a smaller neighborhood of w_0 has positive measure and base at least some $r' > r$. The ratio of outside mass to that inside mass is at most a constant times $(r/r')^{2m}$. The uniform multiplicative comparison transfers this concentration to the normalized measures in (1).

For fixed positive a_i, a_j , the function $(a_i t + a_j w_0)^{-1}$ is bounded and continuous on $(0, \infty)$. Dividing (1) by its positive common mass and taking $m \rightarrow \infty$ therefore gives comparison of

$$\sum_{i,j} \frac{|K_{ij}|^2}{w_0(a_i + a_j)} = \frac{g_A(K)}{2w_0}.$$

There are finitely many eigenvalues for each channel. The original reference was arbitrary. This proves complete SLD dominance. $\square$

1.2 Entropic and testing instances

Theorem 2 (Petz Rényi, Umegaki and Chernoff comparison). *For arbitrary finite-dimensional channels, the following complete orders on faithful state pairs and arbitrary finite untouched references coincide:*

1. SLD dominance;
2. unrestricted Umegaki relative-entropy dominance;
3. dominance for any one fixed standard Petz Rényi divergence D_α , for each $0 < \alpha < 2$;
4. dominance for the quantum Chernoff divergence $C(A, B) = -\log \min_{0 \leq u \leq 1} \text{Tr } A^u B^{1-u}$.

The statement concerns standard Petz Rényi divergences, not sandwiched Rényi divergences. It asserts equality of qualitative complete preorders, not equality of numerical contraction coefficients.

Proof. For $\alpha < 1$, order for D_α is order for the operator-convex generator $-x^\alpha$; for $1 < \alpha < 2$ it is order for x^α . These generators are finite at zero and have nonzero interior resolvent mass. For $\alpha = 1$ the generator is $x \log x$, also in this class. Theorem 1 proves their equivalence to SLD.

SLD dominance compares every Petz D_u for $0 < u < 1$, hence compares the overlaps $Q_u = \text{Tr } A^u B^{1-u}$ in the direction $Q_u(P) \leq Q_u(Q)$. Taking minima gives $C(P) \geq C(Q)$. Conversely, append an untouched classical flag and symmetrize a faithful pair:

$$\tilde{A} = \frac{1}{2}(A \otimes |0\rangle\langle 0| + B \otimes |1\rangle\langle 1|), \quad \tilde{B} = \frac{1}{2}(B \otimes |0\rangle\langle 0| + A \otimes |1\rangle\langle 1|).$$

Then $Q_u(\tilde{A}, \tilde{B}) = (Q_u(A, B) + Q_{1-u}(A, B))/2$ is symmetric and convex in u ; convexity follows from its spectral sum of exponentials. Its minimum occurs at $u = 1/2$. Trace cyclicity gives

$$C(\tilde{A}, \tilde{B}) = -\log Q_{1/2}(A, B) = D_{1/2}(A\|B)/2.$$

Complete Chernoff order therefore implies complete $D_{1/2}$ order, and hence SLD order. The flag is finite and untouched. $\square$

1.3 Why this covers the whole finite-at-zero operator-convex class

The standard representation theorem [2, Theorem 7.1] writes a continuous operator-convex function on $[0, \infty)$ as

$$f(x) = f(0) + a_0x + cx^2 + \int_{(0, \infty)} \left(\frac{x}{1+s} - \frac{x}{x+s} \right) d\mu(s), \quad \int \frac{d\mu(s)}{(1+s)^2} < \infty.$$

The elementary identity

$$\frac{x}{1+s} - \frac{x}{x+s} = \frac{x-1}{(1+s)^2} + \frac{s}{(1+s)^2} \frac{(x-1)^2}{x+s}$$

puts it in Theorem 1's form, with $d\nu(s) = s(1+s)^{-2}d\mu(s)$. Both required integrability bounds follow immediately. Conversely, that form has finite value at zero and is operator convex. Thus, modulo affine terms, exactly three cases occur: zero; a positive pure quadratic; or nonzero interior resolvent mass (with or without a quadratic part). The proof does not require the measure to have a density or infinitely many support points. A single resolvent atom already belongs to the last case. This is a classification for a fixed chosen generator; it is not an algorithm for recognizing a generator from an arbitrary function oracle.

2 The quadratic endpoint admits finite compression

Write

$$\mathcal{G}_A(K) = \text{Tr } KA^{-1}K, \quad D_2(A||B) = \log \text{Tr } A^2B^{-1}.$$

All inverses are on the effective output support. This is standard Petz Rényi order two, not the sandwiched divergence or an unspecified quantum χ^2 . The associated quadratic generator is $(x-1)^2$.

Theorem 3 (Explicit finite reference cutoff). *Let P, Q be finite-dimensional channels with common input, and let b be the effective output dimension of Q . If complete Petz D_2 dominance of P over Q fails on faithful pairs, it fails with a reference of dimension at most $1+b^2$. For POVMs E, F , if the target F has v nonzero outcomes, the bound improves to $1+v$. Hence complete D_2 dominance of rational complex strictly positive POVMs is decidable by a total exact algorithm.*

Proof. First, for a faithful state ρ and traceless Hermitian H ,

$$\text{Tr}[\Phi_R(\rho + tH)]^2[\Phi_R(\rho)]^{-1} = 1 + t^2 \mathcal{G}_{\Phi_R(\rho)}(\Phi_R(H)). \quad (2)$$

The linear term is $2t \text{Tr } H = 0$ and there is no remainder. Conversely every faithful pair is of this form with $\rho = \sigma$, $H = \alpha - \sigma$, $t = 1$. Since logarithm is increasing, complete D_2 order is exactly complete comparison of these RLD quadratic forms.

Suppose the quadratic comparison fails at a faithful input R and Hermitian traceless tangent H on $\mathbb{C}^a \otimes \mathbb{C}^n$. Put

$$W = \text{Tr}_a R > 0, \quad \tilde{R} = (I \otimes W^{-1/2})R(I \otimes W^{-1/2}), \quad h = (I \otimes W^{-1/2})H(I \otimes W^{-1/2}).$$

Then $\tilde{R} > 0$, $\text{Tr}_a \tilde{R} = I_n$, and $h = h^*$. For $\Phi = P, Q$ put $A_\Phi = \Phi_R(\tilde{R})$ and $K_\Phi = \Phi_R(h)$. Reference congruence commutes with the channels. Put $C = I \otimes W^{1/2}$. The exact cancellation identity

$$\text{Tr}(CK_\Phi C)(CA_\Phi C)^{-1}(CK_\Phi C) = \text{Tr } W \text{Tr}_{\text{out}}(K_\Phi A_\Phi^{-1}K_\Phi)$$

shows that the original violation is $\text{Tr } W\mathcal{D} < 0$, where

$$\mathcal{D} = \text{Tr}_{\text{out}(P)} K_P A_P^{-1} K_P - \text{Tr}_{\text{out}(Q)} K_Q A_Q^{-1} K_Q.$$

Consequently some unit reference vector ξ satisfies $\langle \xi, \mathcal{D}\xi \rangle < 0$.

Fix an orthonormal basis $(e_c)_{c=1}^b$ of the target effective output and set

$$z_c = A_Q^{-1} K_Q(e_c \otimes \xi).$$

Let S span ξ and every reference component of these b vectors. Each has b reference components, so $d = \dim S \leq 1 + b^2$. Choose an isometry $V : \mathbb{C}^d \rightarrow \mathbb{C}^n$ onto S , and compress the entire physical baseline and tangent:

$$\tilde{R}' = (I_a \otimes V^*) \tilde{R} (I_a \otimes V), \quad h' = (I_a \otimes V^*) h (I_a \otimes V).$$

This preserves faithfulness, Hermiticity, and $\text{Tr}_a \tilde{R}' = I_d$. For either output, A'_Φ and K'_Φ are the corresponding common reference compressions. Write $\xi' = V^* \xi$.

Every z_c lies in $\mathbb{C}^b \otimes S$. Therefore

$$A'_Q(I_b \otimes V^*) z_c = (I_b \otimes V^*) K_Q(e_c \otimes \xi) = K'_Q(e_c \otimes \xi').$$

Thus the compressed target inverse problem has exactly the compressed solution, and each target scalar quadratic term is preserved. For a source basis vector, use

$$\langle u, A^{-1}u \rangle = \sup_z \{2 \text{Re} \langle u, z \rangle - \langle z, Az \rangle\}. \quad (3)$$

Compression restricts this supremum to the source output tensored with S , so each source term can only decrease. Summing the partial-trace basis terms gives $\langle \xi', \mathcal{D}' \xi' \rangle < 0$.

For POVMs the output matrices are block diagonal. A target inverse solution for outcome j occupies only its own classical block and has the one reference component $z_j = (r_j^F)^{-1} k_j^F \xi$. Hence $S = \text{span}\{\xi, z_1, \dots, z_v\}$ suffices, and $d \leq 1 + v$.

To restore a physical scalar witness, choose sufficiently small $\eta > 0$ and the faithful reference state

$$W_\eta = (1 - \eta) |\xi'\rangle \langle \xi'| + \eta I_d/d.$$

Then $\text{Tr } W_\eta \mathcal{D}' < 0$. Undo the congruence:

$$R' = (I_a \otimes W_\eta^{1/2}) \tilde{R}' (I_a \otimes W_\eta^{1/2}), \quad H'_0 = (I_a \otimes W_\eta^{1/2}) h' (I_a \otimes W_\eta^{1/2}).$$

The matrix R' is faithful and has trace one. Set $H' = H'_0 - (\text{Tr } H'_0) R'$. This is traceless and preserves the difference, since

$$\mathcal{G}_A(K - cA) = \mathcal{G}_A(K) - 2c \text{Tr } K + c^2 \text{Tr } A$$

and both channels preserve trace. For small nonzero t , $R' + tH'$ is also a faithful state; (2) gives the required finite-reference state-pair violation.

Finally fix a reference dimension $d \leq 1 + v$. Existence of a violation is a first-order real-algebraic sentence in the real and imaginary coordinates of $R' > 0$, $H' = H'^*$, $\text{Tr } R' = 1$, $\text{Tr } H' = 0$, and output inverse variables. Introduce $T_i M_i(R') = I_d$ for each output block. Strict positivity of the baseline and nonzero effects ensure the inverses exist. Negativity of the source-minus-target quadratic form is a polynomial inequality. Rational complex effects give rational real coefficients. Real quantifier elimination decides the sentence. Testing the finitely many dimensions $1, \dots, 1 + v$ therefore decides complete order. No efficiency claim is made. $\square$

The theorem is independent of the nonlocal-game reduction. It needs no condition-number promise, no free Positivstellensatz, and no substitution of arbitrary operator tuples for physical reference states. The mathematical bound also holds for nonzero, possibly singular POVM effects because a faithful physical-reference input makes each nonzero measurement block invertible. The effective-support convention handles general channels.

Corollary 1 (Explicit unconditional endpoint separation for general channels). *Complete Petz D_2 order is strictly weaker than complete SLD order on finite channels, even on rational channel data. This separation does not assume Dots’ hardness theorem.*

Proof. For a POVM with invertible effects, define the efficient instrument $\mathcal{I}_E(\rho) = \bigoplus_i \sqrt{E_i} \rho \sqrt{E_i}$. For any finite reference put $C_i = \sqrt{E_i} \otimes I_R$. Since $\sum_i C_i^2 = I$, cyclicity gives, for faithful α, β ,

$$\sum_i \text{Tr}(C_i \alpha C_i)^2 (C_i \beta C_i)^{-1} = \sum_i \text{Tr} C_i^2 \alpha \beta^{-1} \alpha = \text{Tr} \alpha^2 \beta^{-1}.$$

Thus $\mathcal{I}_E$ and the identity channel are completely D_2 -equivalent. Take the rational channel

$$\mathcal{I}_E(\rho) = \frac{1}{5} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \rho \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \oplus \frac{1}{5} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \rho \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

It is the instrument for $E_{\pm} = (I \pm \frac{3}{5}Z)/2$ and has rational superoperator and Choi entries. At $\rho = I/2$, $H = X/2$ the identity channel’s SLD information is 1. Each instrument block contributes $8/25$, hence the total is $16/25 < 1$. Therefore $\mathcal{I}_E$ does not SLD-dominate the identity. The SLD-to-quadratic implication was already proved, so this is strictness. $\square$

This example was developed in the operational investigation and independently checked in the endpoint investigation. It also falsifies a proposed route from instrument preservation of RLD information to relative-entropy preservation. The analogous strict separation with *both channels POVMs* is still obtained here only conditionally from Dots’ hardness theorem.

3 The resulting classification and its exact dependency

Corollary 2 (Finite-at-zero classification). *Fix a continuous operator-convex generator on $[0, \infty)$, and compare its standard modular Petz divergence after two channels, on every faithful state pair and every finite untouched reference. Modulo affine terms:*

1. *If the generator is zero, comparison is trivial.*
2. *If it is a positive pure quadratic, comparison has the finite cutoff of Theorem 3; it is decidable for rational strictly positive POVMs.*
3. *If it has nonzero interior resolvent mass, comparison is exactly complete SLD order by Theorem 1.*

Assuming Dots’ claimed complete-SLD hardness reduction is correct, the third case is coRE-complete on the identical rational strictly positive POVM input class. Cases 1–3 exhaust the stated generator class.

Proof. The representation and its affine normalization were proved above. Affine divergences give the same constants at both outputs. A positive quadratic multiple induces precisely D_2 order, and all remaining functions have nonzero interior measure. The identity map on the two input POVM lists transfers the manuscript’s SLD predicate to each fixed generator in case 3. Membership and hardness are inherited from that equivalent predicate; no numerical evaluation or effective reconstruction of the fixed representing measure is needed. This last conclusion assumes the named hardness theorem, whereas the classification of preorders does not. $\square$

In particular, on rational strictly positive POVMs,

$0 < \alpha < 2$: coRE-complete, conditionally;	$\alpha = 2$: decidable, independently.
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This refers to *each fixed standard Petz Rényi parameter*. It is not a promise problem in which the algorithm is supplied an arbitrary real parameter. The Umegaki and Chernoff instances induce the same complete preorder.

An arbitrarily small fixed resolvent component changes the decision boundary. For any fixed rational $\varepsilon > 0$, let

$$f_\varepsilon(x) = (x - 1)^2 + \varepsilon \frac{(x - 1)^2}{x + 1}.$$

Its complete order is SLD order, whereas f_0 has the finite cutoff above. The rational instrument example explicitly separates these two orders for every $\varepsilon > 0$, without invoking the hardness reduction. This is a qualitative, universally quantified order statement. It does not give a uniform numerical separation as $\varepsilon \downarrow 0$ and does not establish fixed-error hardness for a new bounded defect.

Finite certificates. For a rational POVM pair, failure of SLD order at a fixed reference dimension is real algebraic and decidable. Strict violations have rational approximants, so unrestricted failure is recursively enumerable. More generally, in an effective hierarchy of decidable fixed levels, a total decision procedure for universal validity is equivalent to a computable sufficient level cutoff: given a decision procedure, output 1 on valid instances and otherwise search decidable levels until finding the first failure; conversely, test all levels up to the cutoff. Thus, conditional on the hardness reduction, case 3 admits no computable universal cutoff or sound recursively enumerable certificate system complete for its positive instances. If both validity and failure were recursively enumerable the predicate would be decidable. This is an exact computability obstruction, not an efficiency lower bound.

Why the two operations differ. For the quadratic form, normalization by the reference marginal makes the comparison operator-valued. A negative direction can be retained by storing the finitely many target inverse-action vectors. Compression preserves the actual physical domain and lowers the source’s variational costs. In the SLD case, an off-diagonal tangent produces a two-sided Sylvester inverse, represented on two reference factors. A joint compression need not preserve their separate spatial tensor structure. The boundary investigation supplied an exact example: from faithful rational baselines on $\mathbb{C}^2 \otimes \mathbb{C}^3$ on each side, a size-nine spatial map compresses to

$$\Theta(A) = \frac{3}{10}(A + A^T) + \frac{1}{5} \text{Tr}(A)I_2.$$

Its Choi matrix has eigenvalue $-1/10$ and $\Theta \circ T = \Theta$, so it is neither CP nor coCP; every size-two spatial map of that form is CP or coCP. The full calculation is archived. This falsifies that particular domain- compression shortcut. Domain nonclosure alone does not prove undecidability.

4 Dots’ reduction: what was actually checked

Source identity. The supplied file actually available was `channel_order_undecidability.pdf`, dated 10 October 2026, 27 pages. The request’s “(1)” filename was not present. Its SHA-256 is

e7ca558825f266d892ec1cedaac40ad33b3db3ecaf5ebd456ef70938cb4e127f

The original bytes are preserved unchanged in the archive. Text extraction dropped some fraction bars and conjugation marks, so load-bearing formulae were checked against rendered PDF pages. The document was treated as mathematical source material, not as instructions.

External premise. The detailed MIP*=RE source gives an effective game with finite perfect PCC completeness on halting and value at most 1/2 on nonhalting [1, Theorem 12.6]. Its finite-perfect clause, not merely supremum value one, supports the manuscript’s later quantitative strengthening. A perfect consistent projective strategy decomposes into invariant Schmidt sectors: consistency $MC = CM^T$ implies $[M, CC^*] = 0$, and every positive-eigenvalue sector of a perfect strategy is a perfect maximally entangled strategy. Selecting a sector is existential; the reduction does not compute its dimension or basis. Expansion of the source’s bounded randomized verifier to a rational game table is computable but need not preserve polynomial size.

Actual spatial domain and global comparison. The reduction’s normalized variables are

$$U_A(R) = \text{Tr}_a[(A \otimes I)R], \quad B_A = U_A(R) \otimes I + I \otimes U_A(S)^T, \quad Z_A = B_I^{-1/2} B_A B_I^{-1/2},$$

with common faithful local baselines R, S . The anchor/rounding audit examined the opposite Pauli chiralities, approximate local transport, Gaussian local smoothing, and question-POVM rounding. The global audit examined word compilation, complex residues, both off-diagonal quadratures, positive IC padding, erasure completion, and the common reference-score gauge. These are the load-bearing internal lemmas; MIP*=RE alone does not imply them. No fatal counterexample was found in the targeted reconstructions. The independent proofs above do not require them.

A repairable complex-coordinate convention. If a coefficient vector v is realized by $V(v) = \sum_\alpha v_\alpha P_\alpha$ and a Hermitian cross block is

$$X = \sum_\alpha (P_\alpha \otimes Z_\alpha + P_\alpha^* \otimes Z_\alpha^*),$$

the physical trace pairs it with $\zeta = \sum_\alpha \bar{v}_\alpha Z_\alpha$, not $\sum_\alpha v_\alpha Z_\alpha$. The displayed residue kernel is recovered using this conjugated coordinate. The source’s surrounding convention must therefore be made explicit/corrected, or equivalently $V(v)$ must be defined with conjugated coefficients. This is a coordinate repair supported by direct calculation, not permission to replace complex residues by real ones. The coefficient in PDF Equation (54) is $8/\lambda$; an extracted “ 8λ ” was an extraction error, not a manuscript error. No source bytes were edited.

Fixed-error strengthening. The quantitative audit reconstructed the perfect maximally entangled sector, the cancellation of unknown strategy dimension in the normalized Fisher gap, the reference-filter entropy margin, its Taylor remainder, and tensor-power amplification. It found no obstruction in Sections 11–14 conditional on the earlier bridge. The computed entropy margin is positive *per game*; tensor amplification converts it to the claimed fixed 1/4 barrier for the manuscript’s bounded defect. This does not prove fixed-error hardness for every new divergence or for ordinary capacities.

Overall source verdict. The imported MIP premise checks; the identified coordinate issue has an explicit repair; the targeted analytic and quantitative audits found no fatal gap. The complete universal game-to-POVM theorem nevertheless remains an internally reconstructed candidate requiring independent specialist or formal validation. All complexity statements in this report name it as a dependency. There is no newly discovered fatal gap used to discard the manuscript, and no claim of unconditional downstream hardness based only on model agreement.

5 Consequences and independent investigations

What the main result adds. The central mathematical mechanism is that an unrestricted finite quantum reference can localize the representing measure of a *single fixed* modular Petz

divergence. This identifies many apparently different complete information preorders, while the explicit quadratic cutoff proves that the endpoint has a different finite-certification mechanism. It answers a structural comparison question rather than adding a list of unrelated hard tasks.

The SLD quantity is the Fisher information of a local state family, and the Petz/Chernoff divergences are distinguishability measures. Equality of their complete channel preorders means that their universal ranking of two devices coincides under the stated reference interface. It does not say that they take the same numerical values, that a reference can be prepared cheaply, or that one channel physically simulates the other. The cutoff gives a finite exact algorithm only at the quadratic endpoint and gives no practical complexity guarantee.

Five Sol investigations and the selection decision.

1. **Stronger undecidability.** A complete conditional construction makes both POVMs informationally complete, equal-trace, and arbitrarily close to a common uniform measurement, with scalar deviations of rank at most four. It gives fixed-additive hardness of a pre-existing relative conditional information coefficient and an instance-dependent robust informative-margin promise. The coefficient is finite per input but unbounded over the input class. This useful robustness result still imports Dots' Sections 3–9, and is not hardness of a bounded $[0, 1]$ SDPI coefficient. Its full construction is archived.
2. **Operational consequences.** A complete zero-rate stabilizer-code argument replaces any finite half-erasure helper by finitely many uses of the same qubit half-erasure channel $\mathcal{S}_2$. Together with established Smith–Yard coding [9] it gives the internally proved characterization $N^c \succeq_{\text{RE}}^c N$ if and only if $P(N \otimes \mathcal{S}_2) = 0$ if and only if $Q(N \otimes \mathcal{S}_2) = 0$. The fixed-helper refinement's priority is unresolved; variable-helper activation is prior work. For genuine channels the receiver and environment must be complementary. The manuscript supplies two chosen POVMs, not such a complementary pair. A complement-preserving compiler is the indispensable missing step for ordinary private/quantum-capacity hardness. No such compiler was developed. Designated- output wiretap statements are a different interface and are kept separate.
3. **Computability boundary.** Exact spatial-domain compression failure, its decomposable matrix-convex hull, and finite independent-source cases were developed. These investigations helped expose why an unrestricted free- positivity shortcut changes the problem. The main advance selected from this direction is the elementary quadratic cutoff proved above.
4. **Foundational transfer.** A standalone two-contraction packing and a conditional fixed six-dimensional Jordan carrier yield a fixed operator system with a hard growing matrix cone. Broad fixed-generator positivity and norm noncomputability already have MIP*-based predecessors [6, 7]. The exact carrier is a specialized sharpening, not a demonstrated resolution of an independent major operator-algebra problem. It also does not give fixed physical input dimension for the eventual POVM channel pair.
5. **Radical invention.** The fixed Petz-parameter proof, exact Chernoff flag, and elementary inverse-action compression closed into the two main theorems. Independent constructions also show that extending the Petz-parameter collapse indiscriminately beyond 2 fails. Root's additional contribution is the general representing-measure localization and the combined classification.

Transfers that were rejected. Strictly positive POVM channels are entanglement breaking: their ordinary quantum and private capacities vanish, and their ordinary classical capacity has a separate finite optimization interface. Thus the source does not give ordinary capacity undecidability merely by changing terminology. No locality-preserving Hamiltonian compiler,

major open-problem resolution, or general theory of all quantum information quantities was produced. Known finite decision/guessing comparison theorems concern stronger experiments and do not supply a finite reference bound for this weaker entropy order. Complete entropy comparison is also not generally CPTP degradation.

6 Falsification and priority assessment

Strongest mathematical attacks. The localization proof was tested against a purely quadratic generator, a single atomic representing measure, diffuse and singular measures, rank-deficient numerators, and tail integrability. The quadratic Hessian is killed exactly by $F_m(0) = 0$, so it cannot falsely produce SLD from the endpoint. The support-point exponential tilt handles singular measures; the stated finite- at-zero condition justifies numerator regularization. Generators infinite at zero remain outside the theorem. Independent fixed-parameter derivations check the signs on both sides of $\alpha = 1$; the Umegaki case uses its own nonzero measure rather than a vanishing power-Hessian formula.

The explicit rational instrument above also establishes strict endpoint separation for general channels without invoking undecidability. The cutoff proof was attacked at normalization, trace gauge, common compression, target preservation, source decrease, and faithful restoration. Its crucial inequality follows from the elementary inverse variational principle, not an unjustified equality after compression. A noncommuting numerical instance compressed reference dimension 11 to 3: target quadratic errors were below 5.7×10^{-14} and both source terms decreased. This is a finite diagnostic only. The general Hessian formula was also checked on 50 random finite matrix instances with maximal relative discrepancy below 8×10^{-16} , and the Chernoff flag on ten instances. Exact symbolic checks support separate domain and packing calculations. None replaces the displayed universal proofs.

Closest primary literature. The 2022 complete less-noisy framework already identifies unrestricted reference relative-entropy comparison and tensorizes it [3]. Hirche’s October 2026 preprint proves the coherent-qubit SLD-to-entropy direction [4, Theorem 3.2, Proposition 3.5]; the reverse from a single preassigned modular Petz generator is the relevant proposed addition here. The earlier Astra operational note proves simultaneous all-generator comparison, whose converse simply selects a resolvent; that is weaker than recovering SLD from each one fixed generator. Hirche–Tomamichel’s coefficient collapse concerns their hockey-stick-integral divergences, a different definition [5, Proposition 4.8]. Petz sufficiency/recovery results compare equality in data processing for one channel, a different quantifier pattern [2]. The endpoint’s qualitative decidability can also be reconstructed from Volčič’s quantitative rational Positivstellensatz [8, Theorem 5.2, Remark 5.3]: normalized positive baselines form a monic free spectrahedron, Hermitian tangents are unrestricted free variables, and the operator-valued RLD difference is rational with positive denominators on the strictly positive POVM domain. Thus a generic computable cutoff is already a consequence of that prior machinery. The elementary $1 + b^2$ and $1 + v$ bounds, and the combined classification, are the more specific potential additions; their exact historical priority remains under review. No matching predecessor for those precise claims has been located in the focused search; that is not proof of originality.

Research allocation and limits. Sixteen distinct Luna Max literature investigations were deployed in organized waves (one repeatedly disconnected), with five Sol Max mathematical investigations. Targeted follow-ups reviewed the load-bearing manuscript and the selected new proofs. Primary-source searches spanned nonlocal games, statistical experiments, finite-reference theorems, capacities, entanglement, resource theories, free positivity, operator algebras, logic, and physics. Only decision-changing findings are reported here; source notes are archived. Worker count is a resource receipt, not evidence of correctness. Searches were finite and do not establish

complete priority coverage. All reference-filter dimensions, coefficient magnitudes, preparation requirements, and game-table expansion costs remain part of the interface; no efficient physical access to the witnesses has been established.

7 Significance and the next indispensable gate

The strongest defensible outcome is a potentially original structural theorem with a sharp computational endpoint, supported by complete internally reconstructed proofs. The classification is broader than a specialist parameter extension: it explains why an entire operator-convex class induces the same complete comparison and why one exceptional endpoint permits finite certification. Conditional on Dots' core, it puts exact decidability and coRE-completeness arbitrarily close in a generator family.

There is currently no basis for rating this a civilization-changing 9–10/10 discovery. The hardness derives from established $MIP^*=RE$ machinery; no new computational power, major independently important conjecture resolution, or capacity/physical no-go transfer has been completed. Priority and independent validation could substantially change the assessment. Even if all claims survive, the evidence presently supports a substantial mathematical research result, not historic significance.

The next indispensable gate is narrow: specialist proof reconstruction of Theorem 1's Hessian/localization and Theorem 3's endpoint, an exact predecessor check, and validation of Dots' shared spatial-domain-to-global-POVM bridge with the conjugation convention corrected. A stronger physical claim additionally needs a complement- or locality-preserving compiler. Those gaps are explicit; they were not filled by inference from shared channels or by numerical checks.

Reconstruction and provenance. The archive contains the unchanged manuscript, five substantive mathematical investigations, primary-literature notes, the root derivation, independent reviews, executable finite diagnostics with results, a SHA-256 manifest, and `ASTRA_HANDOFF.json`. Repositories were read at [Astra revision 902172b6](#) and [openai/math revision fd4aeeb2](#). No scientific claim was published and no commit, push, or pull request was made.

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